



Exercise training does not affect heat tolerance in Chinook salmon (*Oncorhynchus tshawytscha*)

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ABSTRACT

The progression of climate warming will expose ectotherms to transient heatwave events and temperatures above their tolerance range at increased frequencies. It is therefore pivotal that we understand species' physiological limits and the capacity for various controls to plastically alter these thresholds. Exercise training could have beneficial impacts on organismal heat tolerance through improvements in cardio-respiratory capacity, but this remains unexplored. Using juvenile Chinook salmon (*Oncorhynchus tshawytscha*), we tested the hypothesis that exercise training improves heat tolerance through enhancements in oxygen-carrying capacity. Fish were trained once daily at 60% of their maximum sustainable swim speed, U_{CRIT} , for 60 min. Tolerance to acute warming was assessed following three weeks of exercise training, measured as the critical thermal maximum (CT_{MAX}). CT_{MAX} measurements were coupled with examinations of the oxygen carrying capacity (haematocrit, haemoglobin concentration, relative ventricle size, and relative splenic mass) as critical components of the oxygen transport cascade in fish. Contrary to our hypothesis, we found that exercise training did not raise the CT_{MAX} of juvenile Chinook salmon with a mean CT_{MAX} increase of just 0.35 °C compared to unexercised control fish. Training also failed to improve the oxygen carrying capacity of fish. Exercise training remains a novel strategy against acute warming that requires substantial fine-tuning before it can be applied to the management of commercial and wild fishes.

1. Introduction

Aquatic animals are being forced to contend with acute increases in water temperature as the frequency and intensity of extreme weather events worsen (Finkel and Katz, 2018; Stillman, 2019). Many environments can nowadays experience extreme heatwave events where temperatures rise upwards of 10 °C above average levels (IPCC, 2014; Perkins-Kirkpatrick and Lewis, 2020). Such extreme heat events will be particularly threatening to ectothermic animals due to the overbearing influence that temperature has on their physiology (Huey and Berrigan, 2001; Huey et al., 2012; Huey and Kingsolver, 1989). An ectotherm's body temperature largely conforms to that of their surrounding environment, as such, ongoing environmental warming will have concomitant impacts on the performance, fitness, and distribution of ectothermic animals (Pörtner and Knust, 2007; Sunday et al., 2011). The progression of climate warming will expose ectotherms to transient heatwave events and temperatures above their tolerance range at increased frequencies (Finkel and Katz, 2018; Stillman, 2019). Therefore, the need to

understand how ectotherms respond to, and perform under, warming temperatures is driving much research in thermal biology (Pinskym et al., 2019; Rodgers et al., 2021a) and, in particular, research relating to organismal heat tolerance (Gomez Isaza et al., 2019; Rodgers et al., 2021b; Sunday et al., 2011).

An ectotherm's capacity to take up oxygen along the oxygen transport cascade is essential for surviving acute warming episodes, like heatwaves (Antilla et al., 2014; Pörtner and Knust, 2007). The relationship between temperature tolerance and oxygen transport capacity has been explored extensively in recent literature (Beers and Sidell, 2011; Ekström et al., 2019; Eliason et al., 2013; Frederich and Pörtner, 2000; Pörtner and Knust, 2007; Sandblom et al., 2016), with some indication that thermal limits are linked on an organism's capacity to deliver oxygen to tissues at elevated temperatures (termed the oxygen- and capacity-limited thermal tolerance hypothesis, OCLTT). Evidence for the OCLTT hypothesis relates to oxygen delivery mechanisms being unable to meet oxygen demands at elevated temperatures, due to functional constraints on cardiac, respiratory, and blood-oxygen

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delivery systems (e.g., haematocrit, haemoglobin concentrations). For instance, haematocrit (which is the proportion of red blood cells to total blood volume) measured following critical thermal maximum (CT_{MAX}) was positively correlated with upper thermal tolerance limits in juvenile Chinook salmon, *Oncorhynchus tshawytscha* (Muñoz et al., 2018), while experimental reductions in haematocrit led to lower heat tolerance in European sea bass, *Dicentrarchus labrax* (Wang et al., 2014). Similarly, levels of circulating haemoglobin were associated with higher thermal tolerance in Antarctic notothenioid fishes (Beers and Sidell, 2011) and, in sockeye salmon *Oncorhynchus nerka*, upper thermal constraints were attributed to physiological limitations in aerobic performance due to cardiac collapse at elevated temperatures (Eliason et al., 2013). However, the OCLTT hypothesis is all but universal (Jutfelt et al., 2018); various studies have shown that the assumptions of the OCLTT hypothesis are not applicable across species (Ern et al., 2015; Gräns et al., 2014; Norin et al., 2014; Poletto et al., 2017), and suggest that mechanisms other than a mismatch in oxygen supply and demand are at play.

Exercise training could have beneficial impacts on organismal heat tolerance through improvements in cardio-respiratory capacity (Farrell et al., 1990; Farrell et al., 1991; Gallagher et al., 2001; Solstorn et al., 2015). Exercise-induced improvements to the cardiovascular system can include cardiac growth and increases in cardiac output (Farrell et al., 1990; Solstorn et al., 2015), increased activity of cardiac enzymes (aerobic enzymes in both spongy and compact myocardium; Castro et al., 2013a; Farrell et al., 1991), and increased protein synthesis in the heart (Houlihan and Laurent, 1987). Haematological changes can also be induced by exercise training (Gallagher et al., 2001; Solstorn et al., 2015; Wells and Weber, 1990); for instance, exercise trained Chinook salmon experienced a 6% increase in haematocrit and a concomitant 15% increase in haemoglobin concentration, which together raised the whole animal blood-oxygen carrying capacity (Gallagher et al., 2001). Wild fish can also undergo “natural training” regimes (e.g., migrations, high-flow environments) that are supported by similar physiological improvements to the cardiovascular system (Dalziel et al., 2015; Eliason et al., 2017; Leonard and McCormick, 1999; Seebacher and Kazerouni-Ghanizadeh, 2021). Such improvements to the cardiorespiratory performance of fish could lend benefits to organismal heat tolerance, but this has yet to be investigated.

Using juvenile Chinook salmon (*O. tshawytscha*), we tested the hypothesis that exercise training would increase acute thermal tolerance limits through enhancements in oxygen-carrying capacity. Chinook salmon were selected as a model because they are highly valuable from commercial, sports fishing, and cultural standpoints, but as a cold-water specialist, Chinook salmon are highly susceptible to acute heatwave events (Crozier et al., 2008; Farrell et al., 2008). Captive and wild populations of Chinook salmon have suffered high levels of mortality following current-day heatwave events across North America, New Zealand, and Australia (Ciric et al., 2019; Daly et al., 2017; Morgan et al., 2019), and are predicted to be vulnerable to increases in temperature associated with climate warming (Crozier et al., 2008; Farrell et al., 2008; Muñoz et al., 2015). Chinook salmon are also active fish and may therefore benefit from exercise training. Chinook salmon were trained once daily for three weeks before acute thermal tolerance was assessed. Acute thermal tolerance is typically measured as the CT_{MAX} , which represents the temperature at which an individual loses their righting capacity and cannot escape local conditions (Becker and Genoway, 1979; Lutterschmidt and Hutchison, 1997; Mora and Maya, 2006). CT_{MAX} data were accompanied by measurements of haematocrit, haemoglobin concentration, relative ventricle size, and relative splenic mass as critical components of the oxygen transport cascade. The effects of exercise training on organismal heat tolerance could have important implications for both aquaculture and for the ecology of wild fishes.

2. Materials and methods

2.1. Fish husbandry

Juvenile Chinook salmon (*O. tshawytscha*, $n = 60$; initial mass = 11.3 ± 2.4 g, total length = 10.4 ± 0.7 cm, mean $\pm$ standard deviation) were sourced from New Zealand King Salmon (Tentburn, New Zealand) and transported to the University of Canterbury in aerated transport bags. Fish were equally distributed among four 48.5 L ($58 \times 38 \times 22$ cm, $l \times w \times h$) tanks within a 1000 L freshwater recirculating aquarium system. Water temperature was maintained at 13 ± 1 °C, and fish were exposed to a 12:12 h light dark cycle. A water temperature of 13 °C was chosen to match the thermal conditions experienced by the fish at the hatchery and because it is within the optimum temperature range for growth in Chinook salmon (Crozier et al., 2010). Fish were fed once daily on a pelleted diet (4–6 mm pellets, EWOS, Cargill Incorporated, Bergen, Norway; same diet being fed at the farm). Fish were allowed to habituate to laboratory conditions for one month prior to experimentation. All experiments were conducted in compliance with the University of Canterbury animal ethics requirements (permit no. 2020/07R).

2.2. Exercise training protocol

Fish were randomly divided into two groups of 30 fish (exercise-trained group and non-trained group; $n = 15$ fish per tank, two replicate tanks per treatment). Exercise training was conducted within a 70 L custom built Blazka-type swimming tunnel (swim chamber dimension: 76×20 cm, $l \times d$) maintained at 13 ± 1 °C. The water flow produced by the swimming tunnel was calibrated using a digital flow meter (Flowtherm NT.2 Höntzsch, Waiblingen, Germany). Fish were netted from their holding tanks and placed into an aerated bucket with water at 13 °C. Fish were then transported within the bucket to a temperature-controlled room where the swimming flume was located, and then net transferred to the swim tunnel. The entire processes lasted approximately 5 min. Fish were allowed 30 min to recover within the swim tunnel prior to training. Control fish were also allowed the same 30 min recovery period. Fish in the exercise-trained group were exercised daily at 60% of their maximum sustainable swim speed, U_{CRIT} (42 cm s^{-1} , 4.1 BL s^{-1} , U_{CRIT} was determined during a pilot experiment, Fig. S1), for 60 min. The water speed in the swim-tunnel was increased gradually ($\sim 10 \text{ cm s}^{-1} \text{ min}^{-1}$) until the desired speed was reached. Fish in the non-trained group remained within the swim tunnel, without flow, for 60 min. Fish were trained in groups (15 fish per group), as previous research indicates no difference in swimming behaviour of fish trained in schools compared to those trained individually (Li et al., 2010b; McFarlane et al., 2001). After training, fish were transferred back into their holding tanks as previously described. Training occurred daily at 9:00–15:00 five days a week for three weeks with the training order randomised. Fish were fed twice daily on a 2% body mass food ration (equivalent to ration provided at the hatchery) at around 8:00 and 15:30.

2.3. Growth rates

Fish body mass (M ; g) and total length (TL; cm) of all fish were measured at the beginning and end of the training period to allow the calculation of growth rates. Fish were lightly sedated using tricaine methanesulfonate (MS-222; 50 mg L^{-1}) buffered with sodium bicarbonate (pH ~ 7) and individually weighed (wet mass) using an electronic balance (Sartorius TE 153D, Göttingen, Germany), TL measured, and then returned to their holding tanks. Tank averages were used to calculate growth rates as the specific growth rate ($\% \text{ day}^{-1}$): $SGR = [(\log(M_F) - \log(M_I)) / t] \times 100$, where M_F and M_I are the final and initial mass, respectively, and t is time (days). Fish body condition factor (K) was calculated as: $K = (M/LT^3) \times 100$.

2.4. Upper thermal tolerance

The upper thermal tolerance of Chinook salmon was assessed following three weeks of exercise training. Upper thermal tolerance was measured as the critical thermal maximum (CT_{MAX}; Becker and Genoway, 1979; Lutterschmidt and Hutchison, 1997). CT_{MAX} trials were conducted within a custom-built water bath (52 × 41.5 × 19.2 cm, 1 × w × h) used to manipulate water temperature. Water bath temperature was monitored using a fluke 51 II handheld digital probe thermometer (Eindhoven, The Netherlands). Water within the water bath was constantly aerated by an airline. Fasted fish were placed within the water bath and allowed to habituate to the water bath for 1 h, after which water temperature was increased at a rate of 0.3 °C min⁻¹ following published protocols (Gomez Isaza et al., 2019; Rodgers et al., 2021b). CT_{MAX} trials were conducted in groups of five fish from the same treatment. CT_{MAX} trials were run over eight trials, totalling $n = 20$ individuals per treatment. Fish were continually observed during CT_{MAX} trials. Loss of equilibrium (LOE), defined as the failure to maintain dorsal-ventral orientation for more than 10 s, was used as the CT_{MAX} endpoint (Lutterschmidt and Hutchison, 1997). Following LOE, blood was immediately sampled (see below) before fish were placed into aerated recovery tanks. Fish were then lightly sedated (MS-222, 50 mg L⁻¹), weighed, measured (TL), and returned to their holding tanks.

2.5. Haematology and organ measurements

A small blood sample was taken from fish via caudal puncture using a 0.3 mL insulin syringe. Blood was sampled from separate groups of unanaesthetised fish at rest (i.e., undisturbed fish; $n = 15$) and immediately following CT_{MAX} (i.e., following LOE; $n = 15$). Blood sampling via caudal puncture can alter blood properties (e.g., spleen translocation, cellular swelling, fluid shifts) as a result of capture-induced stress (Marino et al., 2001). However, previous work has shown that rapidly obtained (collected within <220 s) caudal blood samples accurately represent blood properties with negligible impacts on haematocrit (H_{CT}), plasma ions, metabolites, and stress hormones (Clark et al., 2011). Blood samples were therefore taken within 2 min (mean = 1:05 min, range = 0:31–1:53 min) of netting (for both post-CT_{MAX} and resting blood samples), placed on ice, and analysed within 1 h of collection. Two blood samples were collected beyond 2 min from capture (both samples were from control fish at rest) and were excluded from the dataset. Blood samples were analysed for H_{CT}, haemoglobin concentration (H_B), and mean corpuscular haemoglobin concentration (MCHC). H_{CT} was measured in 75 mm capillary tubes after spinning blood for 3 min in a microcentrifuge (Haemofuge, Haraeus Sepatech, USA) at 12,000 g. H_{CT} was calculated as the proportion of red blood cells in whole blood. H_B of whole blood was quantified spectrophotometrically at 400 nm using a Sigma-Aldrich haemoglobin assay kit (MAK115; St. Louis, USA). Blood samples were analysed in duplicate. MCHC was calculated as $H_B \times 100 / H_{CT}$. A subset of fish ($n = 15$) from each treatment were euthanised with an overdose of MS-222 (250 mg L⁻¹) for organ measurements. The ventricle and spleen were dissected from fish and individually weighed to obtain wet mass (g) and expressed as a relative measure in terms of per cent body mass (relative ventricular [RVM] and splenic mass [RSM], respectively).

2.6. Statistical analyses

Statistical analyses were performed in the R programming environment using the RStudio interface (version 1.2.1335). All variables were tested for normality and homoscedasticity with Shapiro-Wilk and Levene tests, respectively. All data were normal and of equal variance. Linear mixed effects models were used to test for statistical differences between control and exercise-trained fish in terms of growth, CT_{MAX}, haematological parameters and organ measurements. Exercise training regime (control or exercise trained) was included as a fixed effect in all

models and tank as a random effect. Body mass was treated as a covariate in all analyses. The *lme* function of the *nlme* package (Pinheiro et al., 2017) was used for all analyses. Post hoc pairwise comparisons were performed using the *emmeans* function of the *emmeans* package (Lenth et al., 2022). Significant differences were accepted as $P < 0.05$. Data are presented as mean ± standard error unless otherwise stated.

3. Results

3.1. Growth

Initial measurements on fish did not differ among the control and exercise-trained groups (initial mass, $F_{1,2} = 0.21$, $P = 0.69$; initial length, $F_{1,2} = 8.39$, $P = 0.10$; initial condition factor, $F_{1,2} = 0.96$, $P = 0.43$; Table 1). Following three weeks of exercise training, the exercised trained fish were, on average, 0.96 ± 0.82 g (mean difference ± standard error) larger than the untrained, control fish. Yet, this difference was not statistically significant ($F_{1,2} = 0.95$, $P = 0.43$). Final length ($F_{1,2} = 3.90$, $P = 0.19$) and condition factor ($F_{1,2} = 1.70$, $P = 0.32$) were also unaffected by exercise training (Table 1).

3.2. Upper thermal tolerance

Chinook salmon in the exercise-trained group (CT_{MAX} = 28.60 ± 0.06 °C, mean ± standard error) tolerated higher temperatures before reaching LOE than salmon in the non-trained group (CT_{MAX} = 28.24 ± 0.08 °C; mean difference in acute thermal tolerance of 0.35 °C, 95% CI = 0.14–0.57 °C; Fig. 1A, B). However, this difference was not statistically significant ($F_{2,36} = 11.2$, $P = 0.08$). The non-trained control group showed greater variation in CT_{MAX} than the exercise-trained group. The CT_{MAX} of control fish ranged between 27.6 and 28.9 °C and had a sample variance of 0.14 °C. In comparison, the CT_{MAX} of exercise trained fish ranged between 28.1 and 29 °C and had a sample variance of 0.08 °C – about half that of the control group. Body mass did not influence the CT_{MAX} ($F_{1,35} = 0.75$, $P = 0.39$) of fish and was therefore removed as a covariate in the final model.

3.3. Haematology and organ measurements

Exercise-training did not affect the H_{CT} ($F_{1,2} = 0.03$, $P = 0.87$), H_B ($F_{1,2} = 1.59$, $P = 0.33$), or the MCHC ($F_{1,2} = 1.57$, $P = 0.34$) of juvenile Chinook salmon relative to control fish (Fig. 2). H_{CT} levels were significantly increased following CT_{MAX} ($F_{1,52} = 61.89$, $P < 0.0001$) relative to blood samples collected at rest in both treatment groups (Fig. 2A). Contrarily, H_B did not differ significantly in resting and post-CT_{MAX} blood samples ($F_{1,52} = 2.35$, $P = 0.13$; Fig. 2B). MCHC was significantly reduced following CT_{MAX} ($F_{1,52} = 10.74$, $P < 0.001$) compared to blood collected from fish at rest (Fig. 2C). Mass significantly influenced H_{CT}

Table 1

Growth performance (mean ± standard error) of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) before and after three weeks of exercise-training. No statistical differences were detected in any metric between the control and exercised groups.

	Control	Exercised
Before		
Mass (g)	11.39 (± 0.27)	11.15 (± 0.71)
Total length (cm)	10.48 (± 0.01)	10.32 (± 0.08)
Condition (K)	0.98 (± 0.01)	1.00 (± 0.02)
After		
Mass (g)	14.37 (± 0.49)	15.33 (± 0.65)
Total length (cm)	11.06 (± 0.04)	11.43 (± 0.26)
Condition (K)	1.04 (± 0.04)	1.01 (± 0.01)
SGR (% day ⁻¹)	0.48 (± 0.15)	0.66 (± 0.01)
Survival (%)	100 (± 0.0)	100 (± 0.0)

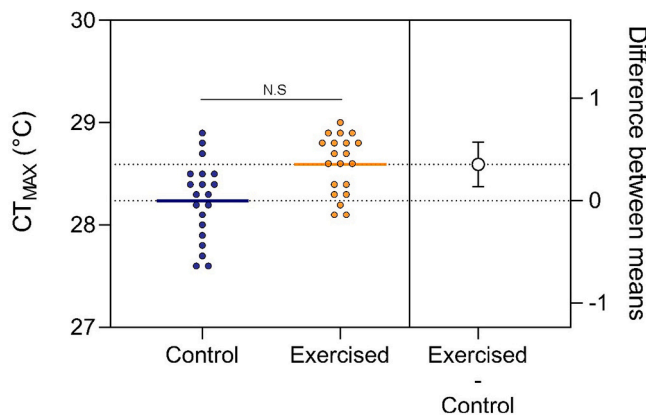


Fig. 1. Critical thermal maximum (CT_{MAX} ; °C) of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) following three weeks of exercise training. Individual data points ($n = 20$) are shown for each treatment group and solid horizontal lines represent the mean of each treatment (left panel). The difference in CT_{MAX} was not statistically significant between the exercised and the control fish. The right panel shows the mean difference (0.35 °C, 95% CI = 0.14 – 0.57 °C) in CT_{MAX} between the exercised and the control groups.

($F_{1, 52} = 8.75$, $P = 0.004$) and H_B ($F_{1, 52} = 6.20$, $P = 0.02$), with larger fish having higher values, and was retained in the final model. Body mass did not influence MCHC ($F_{1, 52} = 0.37$, $P = 0.54$). Three-weeks of exercise training did not influence the ventricular mass ($F_{1, 2} = 3.03$, $P = 0.22$) or splenic mass ($F_{1, 2} = 3.27$, $P = 0.21$) of Chinook salmon compared to untrained, control fish (Fig. 3).

4. Discussion

The increasing frequency and intensity of acute heatwaves events are set to imperil global biodiversity as upper thermal limits are surpassed. It is therefore pivotal that we understand species' physiological limits and the capacity for various controls to plastically alter these thresholds. Here, we hypothesised that exercise training, which can occur in natural and captive settings, would increase organismal tolerance to acute warming through enhancements in oxygen-carrying capacity. In contrast to our hypothesis, we found that three-weeks of exercise training was insufficient to improve tolerance to acute warming in juvenile Chinook salmon, causing only a small increase in upper thermal

limits (measured as CT_{MAX}). Examinations into the oxygen carrying capacity (blood and organ level measurements) of exercise-trained fish showed no changes compared to non-exercised control fish. These data provide springboard evidence for future research into how exercise training could be harnessed as a tool to increase organismal tolerance to abiotic stressors.

Exercise-trained fish attained a CT_{MAX} that was marginally, but not significantly, higher than control fish. Fish upregulate blood oxygen carrying capacity following periods of stress (Franklin et al., 1991; Gomez Isaza et al., 2021), as evidenced by higher haematocrit and haemoglobin concentrations following CT_{MAX} compared to blood samples collected at rest; yet this effect was not enhanced by exercise training. The lack of an effect of exercise training on heat tolerance may be explained by the moderating effect of temperature (McClelland et al., 2006; Pang et al., 2013; Seebacher and Kazerouni-Ghanizadeh, 2021). For example, acclimation to water flow improved the swimming performance of mosquitofish (*Gambusia holbrooki*) at low water temperatures compared to fish maintained under no-flow conditions, but the effect was not seen as fish were pushed towards critical warm temperatures (Seebacher and Kazerouni-Ghanizadeh, 2021). Similarly, juvenile

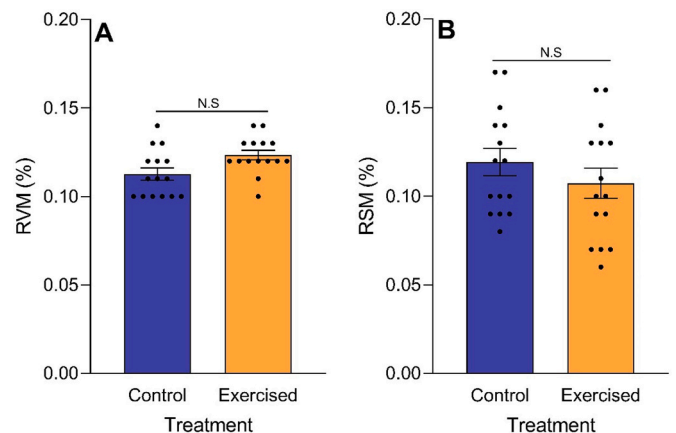


Fig. 3. (A) Relative ventricular (RVM) and (B) splenic masses (RSM; % body mass) of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) following three weeks of exercise training. Data are presented as mean $\pm$ standard deviation and solid circles represent individual data points ($n = 15$). Exercise training did not affect either variable (represented by the N.S.; not significant).

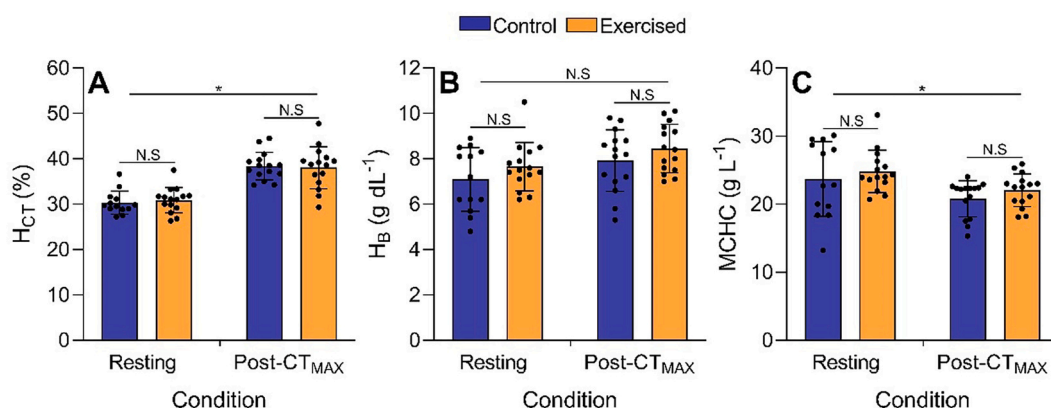


Fig. 2. Haematological metrics of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) following three weeks of exercise training collected under two conditions: at rest or after critical thermal maximum measurements (post- CT_{MAX}). (A) Haematocrit (H_{CT} ; % red blood cells), (B) haemoglobin concentration (H_B ; g dL⁻¹), and (C) mean corpuscular haemoglobin concentration (MCHC; g dL⁻¹). Blue bars represent control (non-trained) fish and orange bars represent exercised fish. Data are presented as mean $\pm$ standard deviation and solid circles represent individual data points ($n = 13$ – 15). Exercise training did not affect blood variables (represented by the N.S.; not significant) under rest or following CT_{MAX} . H_{CT} levels were significantly elevated following CT_{MAX} compared to blood samples collected at rest in fish from both treatment groups, as indicated by an asterisk (*), while MCHC was reduced post- CT_{MAX} in fish from both treatment groups. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

qingbo (*Spinibarbus sinensis*) acclimated to low temperatures (15 °C) experienced the beneficial effects of exercise training (improved swimming performance, aerobic scope), but these benefits were not seen in fish at warm acclimation temperatures (25 °C; Pang et al., 2013). These results suggest that acclimation temperature can impact the effectiveness of exercise training in fish. The mechanistic explanations for the lack of these effects at elevated temperatures have been attributed to aerobic metabolic capability (e.g., arterial oxygen delivery, a decrease in dissolved oxygen) and ion fluxes being compromised at elevated temperatures (Farrell et al., 1996; Gonzalez and McDonald, 1994; Pang et al., 2013). Jain and Farrell (2003) found that the repeat swimming performance of rainbow trout (*Oncorhynchus mykiss*) was compromised at warm, but not cold, temperatures, with warm-acclimated fish failing to attain a similar swimming speed on a second U_{CRIT} swim after a 40 min recovery period. Plasma lactate concentrations were much greater in warm acclimated fish prior to the second U_{CRIT} trial that surpassed a lactate threshold and hindered performance (Jain and Farrell, 2003). It is unlikely that lactate levels were driving the lack of an effect on upper thermal tolerance in our study, but the null effect may be linked to other variables equally affecting control and exercise-trained fish at elevated temperatures (e.g., decrease in dissolved oxygen, ion channel and neural imbalance, protein or enzyme limitation, effects on membrane fluidity; Clark et al., 2008; Farrell et al., 1996; Gonzalez and McDonald, 1994; Iftikar and Hickey, 2013; Overgaard et al., 2012). However, future work would benefit from measuring lactate and glucose levels of fish following exercise training and CT_{MAX} to help differentiate the anaerobic and aerobic components of these tests.

More simply, it is possible that exercise training does not confer tolerance to warming temperatures. A recent study found that exercise training increased the average maximum heart rate during an acute warming challenge in rainbow trout but did not affect peak heart rate nor the thermal tolerance parameters of cardiac function (measured as the Arrhenius break point temperature of maximum heart rate and temperature where cardiac arrhythmias appeared), that is despite large increases in haematological parameters (Papadopoulou et al., 2022). These data are congruent with our findings, which together show that cardiac nor whole animal thermal tolerance are affected by exercise training in salmonids.

Exercise training can induce a series of physiological changes (see various reviews Davison, 1997; McKenzie et al., 2021; Palstra and Planas, 2011). Yet, physiological changes following periods of exercise training can be small and context dependent. For instance, some studies have shown beneficial improvements to the cardiorespiratory system following prolonged exercise training (Gallaugh et al., 2001; Papadopoulou et al., 2022; Solstorn et al., 2015; Wells and Weber, 1990), while others have seen no such changes (Khan et al., 2014; Thorarensen et al., 1993; Young and Cech, 1993). Some argue that training at low water speeds, well below U_{CRIT} , is most beneficial because swimming at low speeds is primarily supported by red muscle (Brown et al., 2011; Davison, 1994). But, Young and Cech (1994) found that swimming at high ($> 3 \text{ BL s}^{-1}$), not low ($0.5\text{--}1.2 \text{ BL s}^{-1}$), water speeds generated beneficial physiological effects in juvenile striped bass (*Morone saxatilis*). These contrasting data highlight the many nuances associated with exercise training and fish physiological improvements. It is therefore not overly surprising that fish in this study did not experience increases in oxygen carrying capacity or heightened heat tolerance given the context-dependency of exercise training on fish.

Here, fish were swum for one hour at 60% of their U_{CRIT} (4.1 BL s^{-1}) for three-weeks. This training regime was chosen because it has proved beneficial in a few fish species (qingbo *Spinibarbus sinensis*, southern catfish *Silurus meridionalis*, darkbarbel catfish *Peltebagnus vachelli*), with authors citing increases in swimming performance, active metabolism, and improvements in stored energy following training (Li et al., 2010a; Li et al., 2010b; Zhao et al., 2012). Several studies also indicate that, at this speed, swimming is sustained aerobically (Burgetz et al., 1998; Lee et al., 2003; Wilson and Egginton, 1994) and so this training regime was

expected to elicit beneficial increases in oxygen carrying capacity. It is possible that longer term training (of months) is required before the beneficial impacts of this training regime are realised. Indeed, we saw an upwards trend in terms of heat tolerance (mean increase of 0.35 °C) following exercise training, accompanied by small increases in haematocrit, haemoglobin concentrations, and ventricular mass. However, none of these parameters were meaningfully increased by the end of the three-week training regime but may translate to larger effect sizes upon longer term training and warrant further investigation. It is also interesting to note that exercise training reduced the variability (reduced range and variance) in CT_{MAX} compared to the control fish – which could be associated with physiological changes at the organ, cellular, and sub-cellular levels. Given the many caveats surrounding exercise training and physiological benefits in fish, a meta-analytical assessment of exercise training and how the many sources of variation (e.g., species, training regime, temperature) contribute to physiological benefits would be most insightful and would help guide the selection of optimal training regimes. Comparative studies using fish species from low-flow (e.g., carp, catfish) versus high-flow (e.g., salmonids) environments could also be most beneficial to help compare the efficacy of exercise training on the cardio-respiratory performance and heat tolerance of fish from contrasting flow lifestyles.

Though we did not find evidence for increased thermal tolerance in exercised-trained Chinook salmon, other work has shown substantial benefits following exercise training across a variety of taxa (Cerqueira et al., 2018; Husak et al., 2015; Zhang et al., 2015). In fish, exercise training has been shown to confer numerous benefits, including improved growth and feed conversion (Khan et al., 2014; Mes et al., 2020; Palstra et al., 2010), increased swimming and muscular capacity (He et al., 2013; Johnston and Moon, 1980; Pang et al., 2013), brain plasticity (Mes et al., 2020), and increased disease resistance (Castro et al., 2013b). Of particular interest, exercise training has been shown to confer cross-protection to various stressors, such as hypoxia (Fu et al., 2011), ultraviolet-B radiation (Ghanizadeh-Kazerouni et al., 2017), and predation stress (Fu, 2015) (Arbeláez-Rojas et al., 2013; Ghanizadeh-Kazerouni et al., 2017), and improve recovery from stress (hauling, handling; Arbeláez-Rojas et al., 2013; Lackner et al., 1988). The mechanisms behind these effects have been linked to the activation of several physiological pathways (e.g., stress response, metabolic pathways, and antioxidant defences; Arbeláez-Rojas et al., 2013; Seebacher and Kazerouni-Ghanizadeh, 2021) following exercise training that facilitate cross-protection to other stressors (Rodgers and Gomez Isaza, 2021). These same mechanisms can hypothetically provide tolerance to acute warming in fish. As such, future research could replicate this experiment using a variety of training regimes and species to begin to untangle how exercise can buffer species against various stressors.

5. Conclusion

Understanding factors that mitigate the impacts of acute warming is a time critical task as climate change exposes species to thermal extremes. Here we show that, contrary to expectations, exercise training did not improve organismal heat tolerance. This result may be limited due to methodological constraints (e.g., training regime, training duration), but presents a way forward by which we can potentially manipulate a fish's flow environment to improve their tolerance to ongoing warming and other stressors. Much research is still required to understand the many nuances associated with exercise training and physiological benefits and warrant further investigations. Exercise training, however, remains a novel management possibility against acute warming that requires substantial fine-tuning before it can be applied.

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Declaration of Competing Interest

The authors declare no competing or financial interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cbpa.2022.111229>.

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